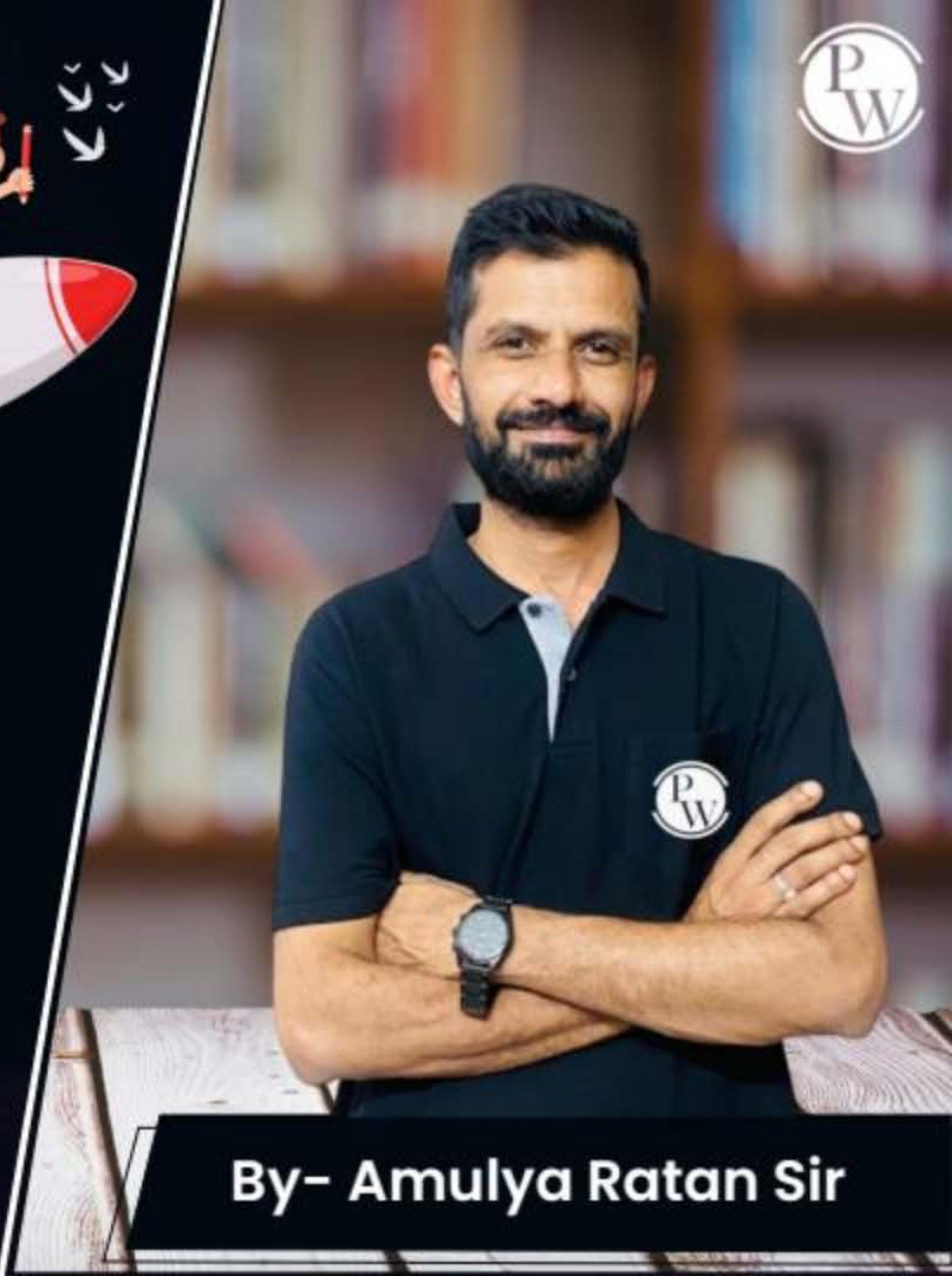
GATE

ALL BRANCHES

General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 03



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Recap of Previous Lecture

Topic Clocks

Time → Angle ?

Topics to be Covered

Topic-1 More on Clocks

Topic-2 Average

Second Pattern:

Angle $\rightarrow$ Time ?

0° OR Coincide:

1 hour → 1 time

~~12 times~~

12 hours → $\boxed{\text{11 times}}$

24 hours → 22 times

180° OR Opposite:

Except
0° & 180°

1 hour ⇒ 2 times

✓ 12 hours ⇒ 11 times

✓ 24 hours ⇒ 22 times

1 hour ⇒ 1 times

~~12 times~~

90° OR Right angle:

Any angle
except 0° & 180°

1 hour → 2 times

~~24 times~~

12 hours ⇒ 22 times

24 hours → 44 times

2-3 @ 90°?

Questionnaire:

$$\text{Time} = \frac{\text{Distance}}{\text{Speed}}$$

$$\boxed{2:27\frac{3}{11}}$$

#Q. In between 2 O' clock and 3 O' clock at what time the hands of clock form 90°?

$$\frac{150^\circ \times 2}{5.5 \times 2} = \frac{300}{11}$$

$$= 27\frac{3}{11} \text{ minutes}$$

Questionnaire:

$4:03\frac{7}{11}$

$4:40$

#Q. In between 4 O' clock and 5 O' clock at what time the hands of clock form 100°?

$$\frac{20^\circ}{5.5} = \frac{40}{11} = 3\frac{7}{11} \text{ minutes}$$

$$\frac{220^\circ}{5.5} = \frac{440}{11} = 40 \text{ minutes}$$

Questionnaire:

1) $6:21\frac{9}{11}$

2) $6:43\frac{7}{11}$

#Q. In between 6 O' clock and 7 O' clock at what time the hands of clock form 60°?

$\frac{120}{5.5} = \frac{240}{11} = \boxed{21\frac{9}{11}}$

$\frac{240}{5.5} = \frac{480}{11} = \boxed{43\frac{7}{11}}$

Questionnaire:

1) $1:23\frac{7}{11}$

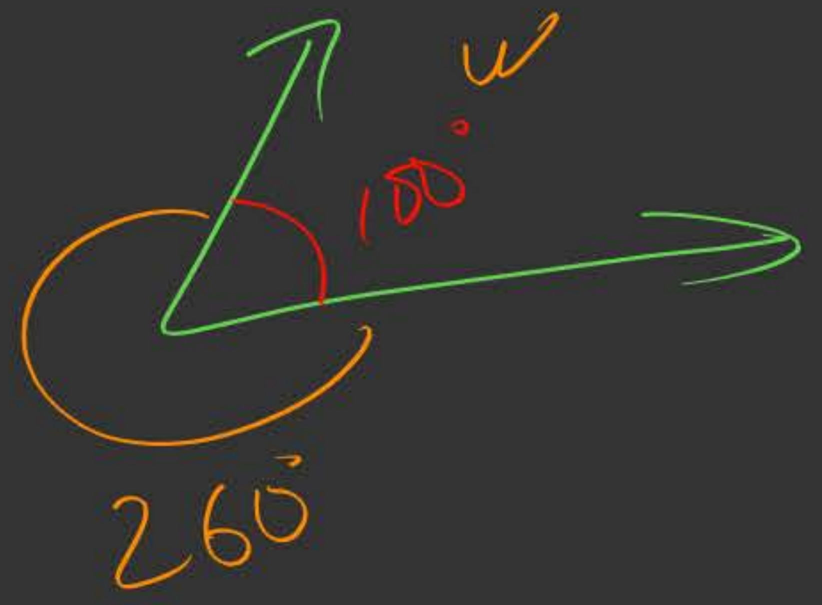
2) $\boxed{1:52\frac{8}{11}}$

#Q. In between 1 O' clock and 2 O' clock at what time the hands of clock form 100°?

$260°$

$$\frac{130°}{5.5} = \frac{260}{11} = 23\frac{7}{11}$$

$$\frac{290°}{5.5} = \frac{580}{11} = 52\frac{8}{11}$$


100°
260°

Questionnaire:

$1\,hr \Rightarrow 2\,times$

Q. 1–2 @ $\underline{\underline{100^\circ}}$

$$\frac{40}{5.5} = \frac{80}{11} = 7\frac{3}{11} \quad \text{back}$$

$(-)$

$1:52\frac{8}{11}$

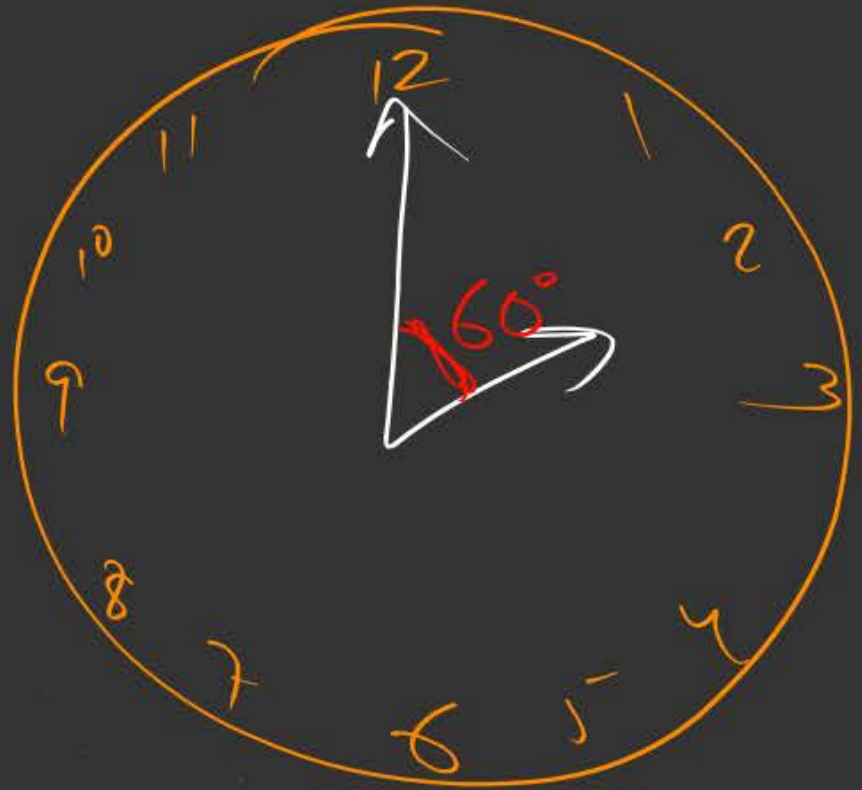
Q

$2:27\frac{3}{11}$

$2-3$ @ 90° $\cancel{270^\circ}$

$$\frac{150^\circ}{5.5} = \frac{300}{11} = 27\frac{3}{11}$$

$$\cancel{\frac{330}{5.5} = \frac{660}{11}} = 60$$



$\cancel{2:60}$

Third Pattern:

2 : 10 → 5

6 : 30 → 15

Gain OR loss

Questionnaire:

+5

1:40

#Q. A Clock which gains 5 minutes in every one hour was set correct at 5am. What would be the time shown by that clock at 1pm the same day?

8 hours

+40 min

Chain Rule

Questionnaire:

#Q. A clock which looses 10 minutes in every one hour was set correct at 4am, what would be the time shown by that clock at 4pm the same day?

−10

2pm

#Q. At what time between 6AM and 7AM will the minute hand and hour hand of a clock make an angle closest to 60°?

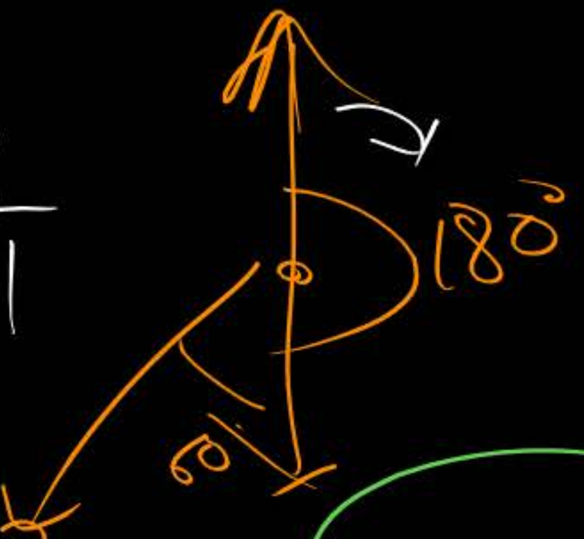
A 6:22 AM

B 6:27 AM

C 6:38 AM

D 6: 45 AM

$$\frac{120}{5.5} = \frac{240}{11} = 21\frac{9}{11}$$



$$\frac{240}{5.5} = \frac{480}{11} = 43\frac{7}{11}$$

$6:21\frac{9}{11}$

$6:43\frac{7}{11}$

#Q. A worker noticed that the hour hand on the factory clock had moved by 225 degrees during her stay at the factory. For how long did she stay in the factory?

$\boxed{\text{Speed of H.H} \Rightarrow 0.5^\circ/\text{min}}$

420 = 7 hrs

$\frac{1}{2}$ $0.5^\circ \rightarrow$ 1 minute

225 $\rightarrow$ 225 × 2

= 450 minutes

7 hrs 30 minutes

A. 3.75 hours

B. 4 hours 15 minutes

C. 8.5 hours

D. 7.5 hours

#Q. It is quarter past three in your watch. The angle between the hour hand and the minute hand is?

3:15

A) 0°

B) 15°

C) 7.5°

D) 22.5°

$3 \to 90^\circ$

$15 \times 5.5 \to 82.5^\circ$

7.5°

AVERAGE

$\boxed{D = S \times T}$ $\boxed{S = \frac{D}{T}}$ $\boxed{T = \frac{D}{S}}$

What ?

Average Speed = 24 km/hr

20 kph

A.M. ← $\frac{\text{Sum}}{\text{No.}}$

'x' km

Home — Office

$$\frac{x}{20} + \frac{x}{30} = \frac{2x}{A.S}$$

30 kph

$$\Rightarrow \frac{5x}{60} = \frac{2x}{A.S}$$

$A.S = 12 \times 2 = 24$ kph

AVERAGE

Equal Distribution

$$\frac{\text{Sum of obs.}}{\text{No. of obs.}}$$

10	15	20
+5		−5
15	15	15

2 mins Summary

Topic Average → Meaning

$$\frac{\text{Sum}}{\text{No.}}$$

Diff b/w A.M & Average

THANK - YOU